

CHAPTER 5 ORDER RELATIONS AND STRUCTURES

Partially Ordered Sets (Posets)

We often use relations to order some or all of the elements of sets. For instance, we order words using the relation containing pairs of words (x, y) where x comes before y in the dictionary. We schedule projects using the relation consisting of pairs (x, y) where x and y are tasks in a project such that x must be completed before y begins. We order the set of integers using the relation containing the pairs (x, y) where $x \leq y$. These types of relations are reflexive, antisymmetric and transitive.

Definition 5.1

A relation R on a set A is called a **partial order** if R is reflexive, antisymmetric and transitive.

The set A together with the partial order R is called a **partially ordered set**, or simply a **poset**.

Notation --- (A, R)

Example 5.1

(a) Let $A = \mathbb{Z}$, are $(\mathbb{Z}, \leq)$ and $(\mathbb{Z}, \geq)$ posets?

Yes. Since the relations “ $\leq$ ” and “ $\geq$ ” are reflexive, antisymmetric and transitive relations on $\mathbb{Z}$, then the relations are partial order on $\mathbb{Z}$.

(b) Let $A = \mathbb{Z}$, are $(\mathbb{Z}, <)$ and $(\mathbb{Z}, >)$ posets?

No. Since the relation “ $<$ ” and “ $>$ ” are not reflexive on $\mathbb{Z}$, then the relations are not partial order on $\mathbb{Z}$.

(c) Let $A = \mathbb{Z}^+$, is $(\mathbb{Z}^+, |)$ a poset?

Yes. Since the relation of divisibility ($x R y \Leftrightarrow x|y$) is reflexive, antisymmetric and transitive relations on $\mathbb{Z}^+$, the relation is a partial order on $\mathbb{Z}^+$.

(d) Let $A = \mathbb{Z}$, is $(\mathbb{Z}, |)$ a poset?

No. Since the relation of divisibility ($x R y \Leftrightarrow x|y$) is not antisymmetric on $\mathbb{Z}$, the relation is not a partial order on $\mathbb{Z}$.

Note

- 1) The most familiar partial orders are relations $\leq$ and $\geq$ on $\mathbb{Z}$ and $\mathbb{R}$. For this reason, in general, a partial order on a set often use the symbols $\leq$ or $\geq$ for R . That is $(A, \leq)$ represents a poset and $a \leq b$ denotes a related to b in a poset.
(Note that the symbol $\leq$ is used to denote **any** poset, not just the “less than or equals” relation).
- 2) We may use symbols such as $\leq_1, \leq', \geq_1, \geq'$ and so on to distinguish partial orders from one another.

Comparable

Definition 5.2

The elements x and y of a poset $(A, \leq)$ are called **comparable**
if $x \leq y$ or $y \leq x$.

When a and b are elements of A such that neither $x \leq y$ nor $y \leq x$, x and y are called **incomparable**.

Note

In partially ordered set, every pair of elements need not be comparable. In Example 5.1(c), the elements 2 and 7 are not comparable, since $2 \nmid 7$ and $7 \nmid 2$. Thus the word “partial” in partially ordered set means that some elements may not be comparable.

Comparable and linearly ordered set

Definition 5.3

If $(A, \leq)$ is a poset and every two elements of A are comparable, then A is called a **linearly ordered set** (or **totally ordered set**) and $\leq$ is called a **linear order** (or **total order**). A linearly ordered set is also called a **chain**.

Example 5.2

Given the poset $(\mathbb{Z}^+, |)$.

- (i) Are the integers 2 and 6 comparable?
- (ii) Are 2 and 3 comparable?
- (iii) Is $(\mathbb{Z}^+, |)$ a chain?

Example 5.3

Let “ $\geq$ ” be the relation “less than or equals”. Is $(\mathbb{Z}, \geq)$ a linearly ordered set?

Note

If $(A, \leq)$ is a poset, we say that $a < b$ if $a \leq b$ but $a \neq b$.

Theorem 5.1

If $(A, \leq_A)$ and $(B, \leq_B)$ are posets, then $(A \times B, \leq)$ is a poset, with partial order $\leq$ defined by

$$(a, b) \leq (a', b') \quad \text{if } a \leq_A a' \text{ and } b \leq_B b'.$$

Lexicographic or dictionary order

Definition 5.4

Let $(A, \leq_A)$ and $(B, \leq_B)$ be posets.

A partial order on $A \times B$, denoted by $\prec$, is defined as follows:

$$(x_A, y_B) \prec (x_A', y_B') \quad \begin{array}{l} \text{if } x_A < x_A' \quad \text{or} \\ \text{if } x_A = x_A' \text{ and } y_B \leq_B y_B'. \end{array}$$

This ordering is called **Lexicographic** or **dictionary order**.

Note

1) If $(A, \leq_A)$ and $(B, \leq_B)$ are linearly ordered sets, then $(A \times B, \prec)$ (the lexicographic order $\prec$ on $A \times B$) is linearly ordered set.

2) Lexicographic ordering is easily extend to Cartesian products $A_1 \times A_2 \times \dots \times A_n$ as follows:

$$\begin{aligned} (a_1, a_2, \dots, a_n) \prec (b_1, b_2, \dots, b_n) & \text{ if and only if} \\ a_1 < b_1 & \quad \text{or} \\ a_1 = b_1 & \quad \text{and} \quad a_2 < b_2 \quad \text{or} \\ a_1 = b_1, a_2 = b_2 & \quad \text{and} \quad a_3 < b_3 \quad \text{or} \\ & \vdots \\ a_1 = b_1, a_2 = b_2, \dots, a_{n-1} = b_{n-1} & \quad \text{and} \quad a_n < b_n \end{aligned}$$

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- 3) Lexicographic ordering is easily extended to compare words or strings of different length. Let $S_1 = a_1a_2\cdots a_m$ and $S_2 = b_1b_2\cdots b_mb_{m+1}\cdots b_n$ where $m \geq n$.
Then $S_1 \prec S_2$ if $(a_1, a_2, \dots, a_m) \prec (b_1, b_2, \dots, b_m)$.

Example 5.4

- (1) Given that $(\{1, 3, 5\}, \leq)$ and $(\{2, 4, 8\}, |)$ are posets.
 $(1, 4) \prec (3, 2)$ because $1 < 3$.
 $(1, 4) \prec (1, 8)$ because $1 = 1$ and $4 \mid 8$.
- (2) Let $A = \{a, b, c, \dots, z\}$ be the ordering alphabet, linearly ordered in the usual way $a \leq b \leq c \leq \dots \leq z$.
Let $A^n = A \times A \times \dots \times A$ be the set of all words of length n .
Then $\text{new} \prec \text{now}$, $\text{park} \prec \text{part}$, $\text{help} \prec \text{hind}$, $\text{jump} \prec \text{mump}$.

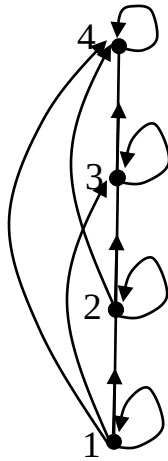
Hasse Diagram

Many edges in the directed graph for a finite poset do not have to be shown since they must be present.

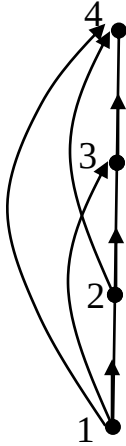
For instance, consider the directed graph for the partial ordering $\{(a, b) \mid a \leq b\}$ on the set $\{1, 2, 3, 4\}$, shown in the following figure (a). Since this relation is a partial ordering, it is reflexive, and its directed graph has loops at all vertices. Consequently, we do not have to show these loops since they must be present; in figure (b) loops are not shown.

Because a partial ordering is transitive, we do not have to show those edges that must be present because of transitivity. For example, in figure (c) the edges $(1, 3)$, $(1, 4)$ and $(2, 4)$ are not shown since they must be present.

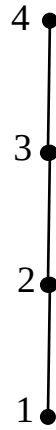
If we assume that all edges are pointed “upward”, we do not have to show the directions of the edges; Figure (c) does not show directions.



(a)



(b)



(c)

Hasse diagram of a poset is a simplified way of representing the digraph of a partial order.

Steps to obtain Hasse diagram of a poset

- 1) Eliminating all cycles of length 1;
- 2) Eliminating all edges that are implied by the transitive property;
(Since a poset is transitive, if $a \leq b$ and $b \leq c$, it follows $a \leq c$. In this case, we draw the edges (a, b) and (b, c) but omit the edge (a, c)).
- 3) Drawing all the edges slanting upwards (initial vertex is below its terminal vertex) and removing all the arrows;
- 4) Replace the circles representing the vertices by dots.

Example 5.5

- (a) Let $A = \{1, 2, 3, 4, 12\}$. Consider the partial order of divisibility on A . That is

$$\forall x, y \in A, x \leq y \Leftrightarrow x \mid y$$

Draw the digraph and the Hasse diagram of the poset $(A, \leq)$.

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- (b) Let D_n denote the set of positive divisors of n .
Draw the digraph and the Hasse diagram of the posets $(D_{24}, |)$ and $(D_{30}, |)$.

Example 5.6

Let $x \leq y \Leftrightarrow x \mid y$ and $x \geq y \Leftrightarrow x$ is a multiple of y .

Draw the digraph and the Hasse diagram of the poset

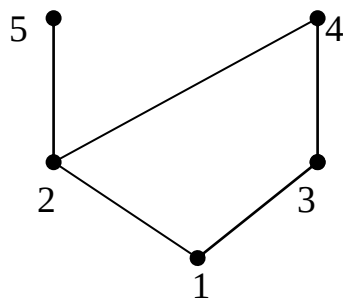
- (a) $(A, \leq)$ for $A = \{1, 2, 4, 8, 16\}$
(b) $(A, \geq)$ for $A = \{2, 3, 4, 5, 15, 60\}$

To recover the digraph of a relation from Hasse diagram, just reverse the instructions given above, using the knowledge that the original digraph was sketched so that all arrows pointed upward:

- (1) Reinsert the direction markers on the arrows making all arrows point upward.
- (2) Add loops at each vertex.
- (3) For each sequence of arrows from one point to a second and from that second point to a third, add an arrow from the first point to the third.

Example 5.7

- (a) Describe the ordered pairs in the relation determined by the Hasse diagram below:



- (b) Determine the Hasse diagram of the relation R on $A = \{1, 2, 3, 4, 5\}$ whose matrix relation is

$$M_R = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Isomorphism

Definition 5.5

Let $(A, \leq_A)$ and $(B, \leq_B)$ be posets and let $f: A \rightarrow B$ be a one-to-one correspondence between A and B .

The function f is called an **isomorphism** from $(A, \leq_A)$ to $(B, \leq_B)$ if for any x and y in A ,

$$x \leq_A y \text{ if and only if } f(x) \leq_B f(y).$$

If $f: A \rightarrow B$ is an isomorphism, we say that $(A, \leq_A)$ and $(B, \leq_B)$ are **isomorphism posets**.

Note

- 1) Two finite isomorphic posets must have the same Hasse diagrams.
- 2) Let $(A, \leq_A)$ and $(B, \leq_B)$ be finite posets and let $f: A \rightarrow B$ be a one-to-one correspondence between A and B , and let H be any Hasse diagram of $(A, \leq_A)$. Then
 - (i) If f is an isomorphism and each label a of H is replaced by $f(a)$, then H will become a Hasse diagram for $(B, \leq_B)$.
 - (ii) If H becomes a Hasse diagram for $(B, \leq_B)$, whenever each label a is replaced by $f(a)$, then f is an isomorphism.

Example 5.8

Let A be the set of all positive integers and let $\leq_A$ be the usual $\leq$ partial order on A . Let B be the set of all positive even integers and let $\leq_B$ be the usual $\leq$ partial order on B .

Is the function $f: A \rightarrow B$ given by $f(x) = 2x$ an isomorphism?

Example 5.9

Let $A = \{1, 2, 3, 6\}$ and $B = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$. Let $(A, |)$ and $(B, \subseteq)$ be the posets. Let the function $f: A \rightarrow B$ be defined by

$$f(1) = \emptyset, \quad f(2) = \{a\}, \quad f(3) = \{b\}, \quad f(6) = \{a, b\}$$

Is f an isomorphism?

Example 5.10

Let $A = \{\emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, \{a, b, c\}\}$ and D_{12} be the set of all positive divisors of 12. Let $(A, \subseteq)$ be the poset with the partial order $\subseteq$ on A and let $(D_{12}, |)$ be the poset with partial order of divisibility on D_{12} .

Are $(A, \subseteq)$ and $(D_{12}, |)$ isomorphic?

Extremal Elements of Posets

Definition 5.6

- (1) An element x in the poset $(A, \leq)$ is called a **maximal element** of A if there is no element a in A such that $x < a$.
- (2) An element y in the poset $(A, \leq)$ is called a **minimal element** of A if there is no element z in A such that $z < y$.

Definition 5.7

- (1) An element I in the poset $(A, \leq)$ is called the **greatest element** of A if

$$a \leq I \quad \forall a \in A.$$

It is denoted by I and often called the **unit element**.

- (2) An element 0 in the poset $(A, \leq)$ is called the **least element** of A if
- $$0 \leq a \quad \forall a \in A.$$

It is denoted by 0 and often called the **zero element**.

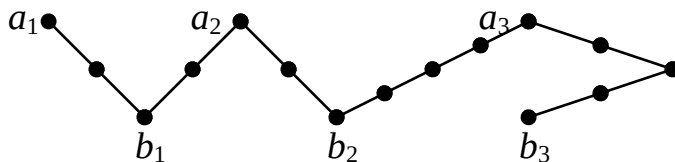
A poset may have any number of maximal element and any number of minimal element. However, the greatest element (or least element) is unique when it exists.

Example 5.11

- (a) Let $(\mathbb{R}^{\text{nonneg}}, \leq)$ be the poset of nonnegative real numbers with the usual partial order $\leq$ (less than or equal to). Then 0 is a minimal element and also the least element of $\mathbb{R}^{\text{nonneg}}$ and there is no maximal element and no greatest element of $\mathbb{R}^{\text{nonneg}}$.
- (b) The poset $(\mathbb{Z}, \leq)$ ($\leq$ means “less than or equal to”) has no minimal element, no least element, no maximal element and no greatest element of $\mathbb{Z}$.

Example 5.12

Consider the poset A whose Hasse diagram is shown in the following figure. The elements a_1 , a_2 and a_3 are maximal elements of A , and the elements b_1 , b_2 and b_3 are minimal elements. Observe that, since there is no line between b_2 and b_3 , we can conclude neither that $b_3 \leq b_2$ nor $b_2 \leq b_3$.



Example 5.13

Find the maximal element(s), minimal element(s), the greatest element and the least element of the following posets.

- (a) $(\{2, 4, 5, 10, 12, 20, 25\}, |)$
- (b) $(\{2, 3, 6, 12, 15, 30\}, |)$
- (c) $(A, \subseteq)$ where $A = P(S)$, $S = \{a, b, c\}$.

Topological sorting

If A is a poset with partial order $\leq$, we sometimes need to find a linear order $\prec$ for the set A that will merely be an extension of the given partial order in the sense that if $a \leq b$, then $a \prec b$. The process of *constructing a linear order* such as $\prec$ is called **topological sorting**. This problem might arise when we have to enter a finite poset A into a computer. The element of A must be entered in some order, and we might want them entered so that the partial order is preserved. That is, if $a \leq b$, then a is entered before b . A topological sorting $\prec$ will give an order of entry of the elements that meets this condition.

The construction of a topological sorting for a general finite poset is based on the fact that *any finite and nonempty poset has a minimal element*. To create a total order for a poset, we follow the following algorithm:

ALGORITHM: TOPOLOGICAL SORTING

procedure *topological sort* (S : finite poset)

$k := 1$

while $S \neq \emptyset$

begin

$a_k :=$ a minimal element of S

{such an element exists since S is finite nonempty}

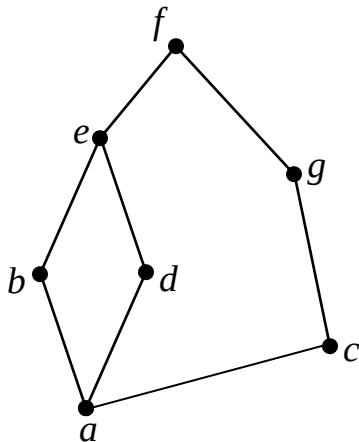
$S := S - \{a_k\}$

$k := k + 1$

end $\{a_1, a_2, \dots, a_n$ is a topological sorting of $S\}$

Example 5.14

Give a topological sorting for the poset whose Hasse diagram is shown below.



The partial order $\prec$ whose Hasse diagram is shown as above is clearly a linear order. it is easy to see that every pair in $\leq$ is also in the order $\prec$, so $\prec$ is a topological sorting of the partial order $\leq$. Note that there are many ways of topologically sorting of a given poset.

Example 5.15

Consider the set $A = \{2, 3, 4, 6, 18, 24\}$ ordered by the “divides” relation. Draw the Hasse diagram of this partial ordering and then give a topological sorting of the partial order.

Topological sorting has an application to the scheduling of projects.

Example 5.16

A development project at a computer company requires the completion of seven tasks. Some of these tasks can be started only after other tasks are finished. A partial ordering on tasks is set up by considering task $X \prec$ task Y if task Y cannot be started until task X has been completed. The Hasse diagram for the seven tasks, with respect to this partial ordering, is shown below. Find an order in which these tasks can be carried out to complete the project.

Greatest lower bound and least upper bound

Definition 5.8

Let $(A, \leq)$ be a poset and let $B \subseteq A$.

- 1) An element $u \in A$ is called an **upper bound** of B if $b \leq u$ for $\forall b \in B$.

An upper bound u_L of B is called the **least upper bound** of B if $u_L \leq$ (all others upper bounds of B). [$u_L = \text{LUB}(B)$]

- 2) An element $l \in A$ is called a **lower bound** of B if $l \leq b \forall b \in B$.

A lower bound l_G of B is called the **greatest lower bound** of B if (all others lower bounds of B) $\leq l_G$. [$l_G = \text{GLB}(B)$]

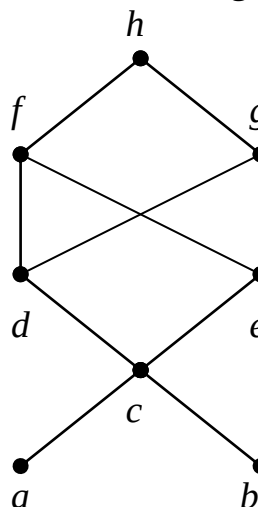
Example 5.17

Let $A = \{a, b, c, d, e, f, g, h\}$ be the poset whose Hasse diagram is shown below.

Find all upper bounds, lower bounds, least upper bound and greatest lower bound of B_1 and B_2 .

- (i) $B_1 = \{a, b\}$

- (ii) $B_2 = \{c, d, e\}$

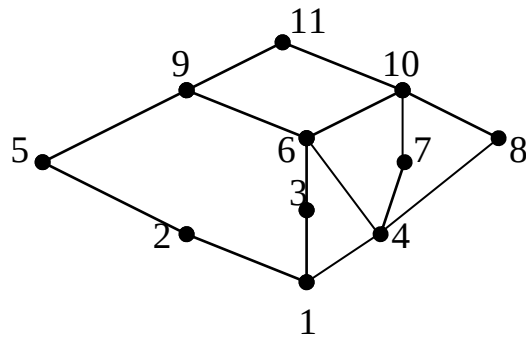


Example 5.18

Let $A = \{1, 2, 3, \dots, 11\}$ be the poset whose Hasse diagram is shown below.

Let $B = \{6, 7, 10\}$. Find the least upper bound and the greatest lower bound of B , if exist.

Exploring all upward paths from
vertices 6, 7, and 10.
Upper bounds of B are _____
 $\text{LUB}(B) =$



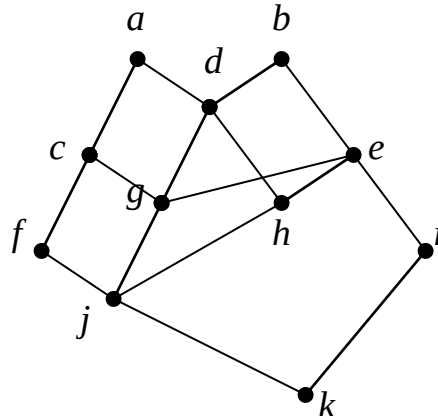
Similarly, by examining all downward paths from vertices 6, 7, and 10.
Lower bounds are _____
 $\text{GLB}(B) =$

Example 5.19

Consider the poset whose Hasse diagram is shown below.

Let $B = \{g, h, i\}$, $C = \{g, h\}$, $D = \{e, f\}$. Find (if it exists)

- | | |
|-----------------------|------------------------|
| (i) $\text{LUB}(B)$ | (ii) $\text{LUB}(C)$ |
| (iii) $\text{LUB}(D)$ | (iv) $\text{GLB}(D)$. |



Lattices

Definition 5.9

A **lattice** is a poset $(L, \leq)$ in which every subset $\{x, y\}$ consisting of two elements has a LUB and GLB.

$\text{LUB}(\{x, y\}) = x \vee y$ (the join of x and y)

$\text{GLB}(\{x, y\}) = x \wedge y$ (the meet of x and y)

Example 5.20

Let n be a positive integer and let D_n be the set of all positive divisors of n . Then D_n is a lattice under the relation of divisibility.

When $n = 20$, $D_{20} = \{1, 2, 4, 5, 10, 20\}$

$$1 \vee i = i \quad \forall i = 2, 4, 5, 10, 20 \quad \text{and} \quad 1 \wedge i = 1 \quad \forall i = 2, 4, 5, 10, 20;$$

$$2 \vee i = i \quad \forall i = 4, 10, 20 \quad \text{and} \quad 2 \wedge i = 2 \quad \forall i = 4, 10, 20;$$

$$2 \vee 5 = 10, \quad 2 \wedge 5 = 1;$$

$$4 \vee 5 = 20, \quad 4 \wedge 5 = 1;$$

$$4 \vee 10 = 20, \quad 4 \wedge 10 = 2;$$

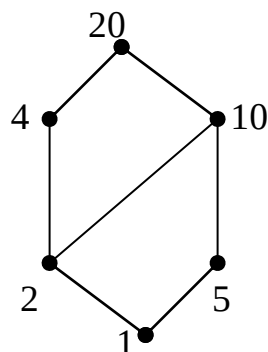
$$4 \vee 20 = 20, \quad 4 \wedge 20 = 4;$$

$$5 \vee 10 = 10, \quad 5 \wedge 10 = 5;$$

$$5 \vee 20 = 20, \quad 5 \wedge 20 = 5;$$

$$10 \vee 20 = 20, \quad 10 \wedge 20 = 10$$

Since every pair of elements has both LUB and GLB, thus $(D_{20}, |)$ is a lattice.



When $n = 30$, $D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$

$$1 \vee i = i \quad \forall i = 2, 3, 5, 6, 10, 15, 30 \quad \text{and} \quad 1 \wedge i = 1 \quad \forall i = 2, 3, 5, 6, 10, 15, 30;$$

$$2 \vee i = i \quad \forall i = 6, 10, 30 \quad \text{and} \quad 2 \wedge i = 2 \quad \forall i = 6, 10, 30;$$

$$2 \vee 3 = 6, \quad 2 \wedge 3 = 1;$$

$$2 \vee 5 = 10, \quad 2 \wedge 5 = 1;$$

$$2 \vee 15 = 30, \quad 2 \wedge 15 = 1;$$

$$3 \vee i = i \quad \forall i = 6, 15, 30 \quad \text{and}$$

$$3 \wedge i = 3 \quad \forall i = 6, 15, 30;$$

$$3 \vee 5 = 15, \quad 3 \wedge 5 = 1;$$

$$3 \vee 10 = 30, \quad 3 \wedge 10 = 3;$$

$$5 \vee i = i \quad \forall i = 10, 15, 30 \quad \text{and}$$

$$5 \wedge i = 5 \quad \forall i = 10, 15, 30;$$

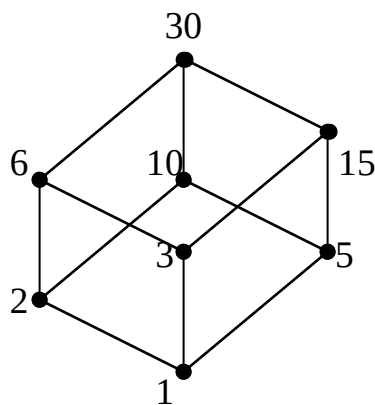
$$5 \vee 6 = 30, \quad 5 \wedge 6 = 1;$$

$$10 \vee 15 = 30, \quad 10 \wedge 15 = 5;$$

$$10 \vee 30 = 30, \quad 10 \wedge 30 = 10;$$

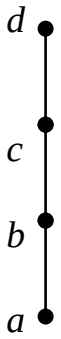
$$15 \vee 30 = 30, \quad 15 \wedge 30 = 15.$$

Every pair of elements has both LUB and GLB, thus $(D_{30}, |)$ is a lattice.

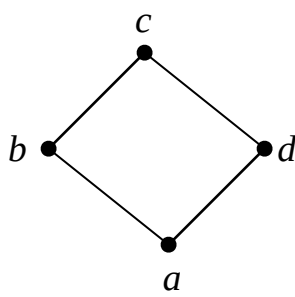


Example 5.21

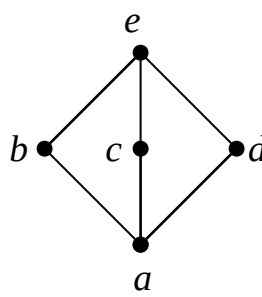
Which of the following Hasse diagrams represent lattices?



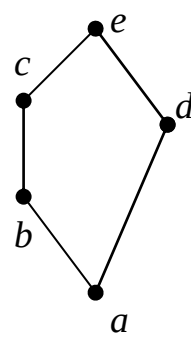
(i)



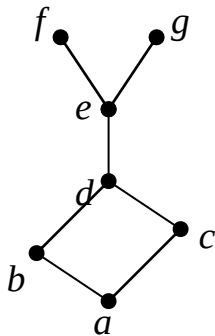
(ii)



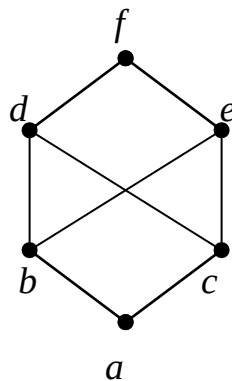
(iii)



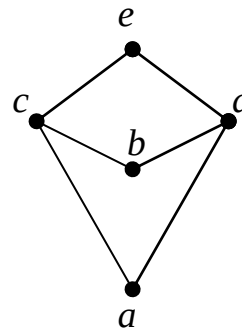
(iv)



(v)



(vi)



(vii)

Hasse diagram (i), (ii), (iii) and (iv) represent lattices

Diagram (v) is not a lattice because $f \vee g$ does not exist.

Diagram (vi) is not a lattice because neither $b \vee c$ nor $d \wedge e$ exist.
Note that $\{d, e\}$ has lower bounds b, c, a but b and c are not comparable.

Diagram (vii) is not a lattice because $c \wedge d$ does not exist.
Note that $\{c, d\}$ has lower bounds a, b but a and b are not comparable.

Sublattices

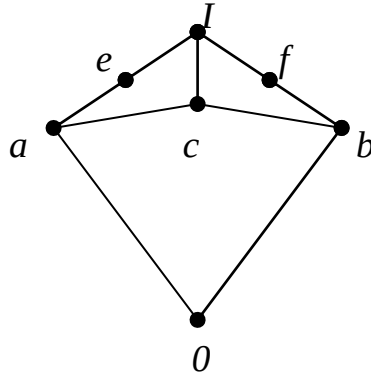
Definition 5.10

Let $(L, \leq)$ be a lattice. A nonempty subset S of L is called a **sublattice** of L if

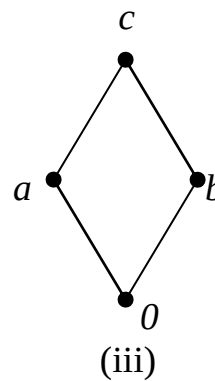
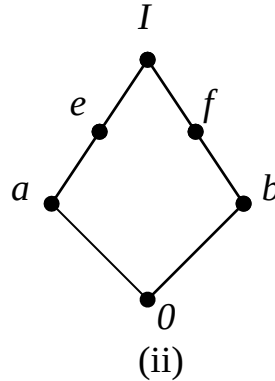
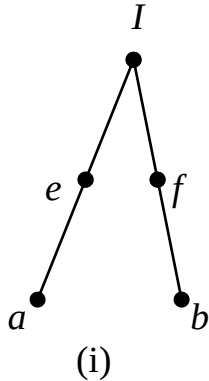
$$x \vee y \in S \text{ and } x \wedge y \in S \text{ whenever } a \in S \text{ and } b \in S.$$

Example 5.22

Consider the lattice L shown below.



Which of the following Hasse diagrams represent sublattices of L ?



The partially ordered subset S_B in figure (i) is not a sublattice of L because

$$a \wedge b \notin S_B \text{ and } a \vee b \notin S_B$$

The partially ordered subset S_C in figure (ii) is not a sublattice of L because

$$a \vee b = c \notin S_C$$

Note that S_C is a lattice when consider as a poset by itself.

The partially ordered subset S_D in figure (iii) is a sublattice of L because

$$x \vee y \in S_D \text{ and } x \wedge y \in S_D \quad \forall x, y \in S_D$$

For example: $a \vee b = c \in S_D$, $a \wedge b = 0 \in S_D$, $a \vee c = c \in S_D$, $a \wedge c = a \in S_D$.

Isomorphic Lattices

If $f: L_1 \rightarrow L_2$ is an isomorphism from the poset $(L_1, \leq_1)$ to the poset $(L_2, \leq_2)$, then L_1 is a lattice if and only if L_2 is a lattice. In fact, if $a, b \in L_1$, then $f(a \wedge b) = f(a) \wedge f(b)$ and $f(a \vee b) = f(a) \vee f(b)$.

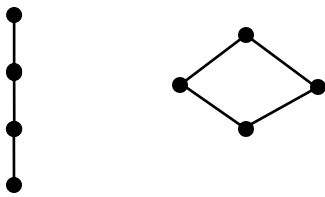
If two lattices are isomorphic, as posets, we say they are **isomorphic lattices**.

Theorem 5.2

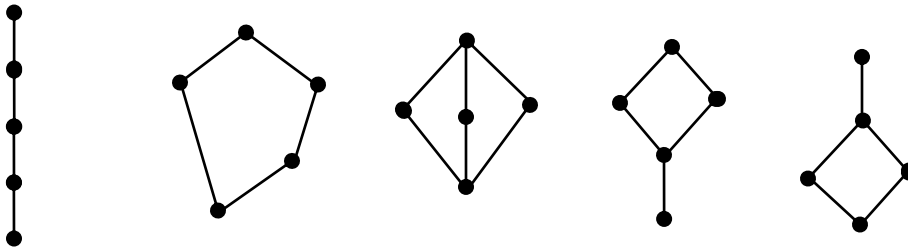
A lattice of order 2 is isomorphic to the 2-element chain.

A lattice of order 3 is isomorphic to the 3-element chain.

A lattice of order 4 is isomorphic to one of these 2:



A lattice of order 5 is isomorphic to one of the following 5 lattices.



Bounded lattices

Definition 5.11

A lattice is said to be **bounded** if it has the least element 0 and the greatest element 1 .

Example 5.23

$(\mathbb{Z}^+, \leq)$ where “ $\leq$ ” means “less than or equals to” is not bounded because it has no greatest element although it has a least element.

Theorem 5.3

Every finite lattice is bounded.

Example 5.24

$(P(S), \subseteq)$ is a bounded lattice.

The greatest element = S because $A \cup S = S, A \cap S = A$.

The least element = $\emptyset$ because $A \cup \emptyset = A, A \cap \emptyset = \emptyset$.

Properties of lattices

$$1) \quad a \wedge b \leq a \leq a \vee b, \quad a \wedge b \leq b \leq a \vee b.$$

$$2) \quad a \vee b = b \Leftrightarrow a \leq b \Leftrightarrow a \wedge b = a.$$

$$3) \quad (i) \quad \text{If } a \leq b, \text{ then } a \vee c \leq b \vee c \text{ and } a \wedge c \leq b \wedge c.$$

$$(ii) \quad \text{If } a \leq b \text{ and } a \leq c \Leftrightarrow a \leq b \wedge c.$$

$$(iii) \quad \text{If } a \leq c \text{ and } b \leq d, \text{ then } a \vee b \leq c \vee d \text{ and } a \wedge b \leq c \wedge d.$$

$$(vi) \quad \text{If } a \leq d \text{ and } b \leq d \Leftrightarrow a \vee b \leq d.$$

$$4) \quad \text{Idempotent Laws:} \quad a \vee a = a, \quad a \wedge a = a.$$

$$5) \quad \text{Commutative Laws:} \quad a \vee b = b \vee a, \quad a \wedge b = b \wedge a.$$

$$6) \quad \text{Associative Laws:} \quad a \vee (b \vee c) = (a \vee b) \vee c$$

$$a \wedge (b \wedge c) = (a \wedge b) \wedge c.$$

$$\therefore \text{LUB}(\{a_1, a_2, \dots, a_n\}) = a_1 \vee a_2 \vee \dots \vee a_n$$

$$\text{GLB}(\{a_1, a_2, \dots, a_n\}) = a_1 \wedge a_2 \wedge \dots \wedge a_n$$

$$7) \quad \text{Absorption Laws:} \quad a \vee (a \wedge b) = a$$

$$a \wedge (a \vee b) = a.$$

$$8) \quad a \vee (b \wedge c) \leq (a \vee b) \wedge (a \vee c)$$

$$a \wedge (b \vee c) \geq (a \wedge b) \vee (a \wedge c).$$

Distributive lattices

Definition 5.12

A lattice L is called **distributive lattice** if for any elements a, b and c in L ,

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c).$$

L is called **nondistributive lattice** if it is not distributive.

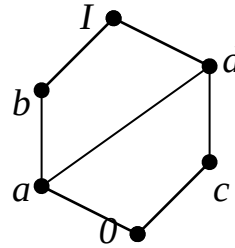
Note

- 1) The distributive law holds when any two elements a, b or c are equal, or when any one of the elements is 0 or I .
Verification of the distributive law is generally a tedious task.
- 2) The lattice $P(S)$ (the power set of S) is distributive because the union $\cup$ and the intersection $\cap$ (the join $\vee$ and the meet $\wedge$, respectively) satisfy the distributive law.

$\therefore$ Any lattice which is isomorphic to a $P(S)$ is distributive.

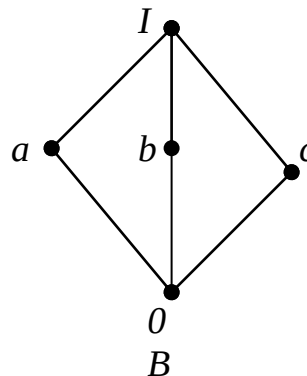
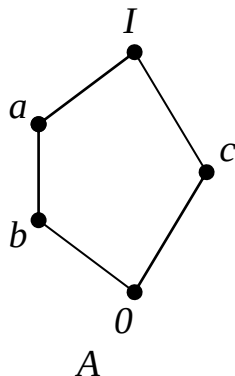
Example 5.25

Is the lattice shown below distributive?



Theorem 5.4

A lattice is nondistributive $\Leftrightarrow$ it contains a sublattice that is isomorphic to one of the two lattices A and B below:



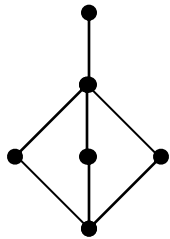
Example 5.26

Show the lattices A and B in Theorem 5.4 are nondistributive.
For lattice A ,

For lattice B ,

Example 5.27

Is the lattices L shown below distributive?



Theorem 5.5

Every chain is distributive.

Theorem 5.6

If L_1 and L_2 are distributive lattices, then $L_1 \times L_2$ is also distributive.

Complement

Definition 5.13

Let L be a bounded lattice with greatest element I and least element 0 and let $a \in L$. An element a' is called a **complement of a** if

$$a \vee a' = I \quad \text{and} \quad a \wedge a' = 0.$$

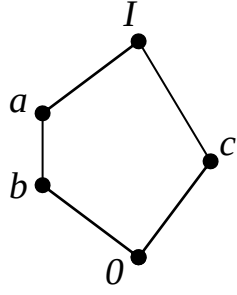
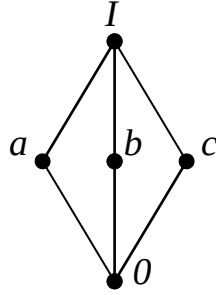
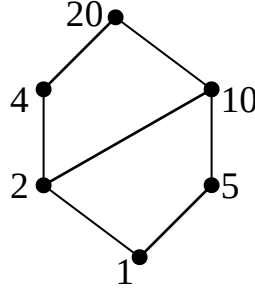
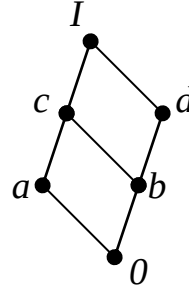
Note that $0' = I$ and $I' = 0$.

Example 5.28

In $(P(S), \subseteq)$, every element $A \subseteq S$ has a complement $\bar{A} = S - A$ because
 $A \cup \bar{A} = S$ and $A \cap \bar{A} = \emptyset$

Example 5.29

Find the complement(s) of each element in the following lattices.

(a) L_a (b) L_b (c) L_c (d) L_d

$$\begin{aligned}
 \text{(a)} \quad & a \vee c = I; \quad a \wedge c = 0 \quad \Rightarrow \quad a' = c, c' = a \\
 & b \vee c = I; \quad b \wedge c = 0 \quad \Rightarrow \quad b' = c, c' = b \\
 & \therefore a' = c, b' = c, c' = a, b, 0' = 1, 1' = 0.
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad & a \vee b = I; \quad a \wedge b = 0 \quad \Rightarrow \quad a' = b, b' = a \\
 & a \vee c = I; \quad a \wedge c = 0 \quad \Rightarrow \quad a' = c, c' = a \\
 & b \vee c = I; \quad b \wedge c = 0 \quad \Rightarrow \quad b' = c, c' = b \\
 & \therefore a' = b, c; \quad b' = a, c; \quad c' = a, b, 0' = 1, 1' = 0
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)} \quad & 1' = 20; \quad 20' = 1 \\
 & 4 \vee 5 = 20; \quad 4 \wedge 5 = 1 \quad \Rightarrow \quad 4' = 5, 5' = 4 \\
 & 2 \text{ and } 10 \text{ have no complement}
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad & a \vee d = I; \quad a \wedge d = 0 \quad \Rightarrow \quad a' = d, d' = a \\
 & b \text{ and } c \text{ have no complement, } 0' = 1, 1' = 0
 \end{aligned}$$

Complemented**Definition 5.14**

A lattice $(L, \leq)$ is called **complemented** if it is bounded and if every element in L has at least one complement.

Example 5.30

Which of the lattices L_a , L_b , L_c and L_d in Example 5.29 are complemented?

Finite Boolean Algebras

Another important kind of partially ordered structure is a Boolean algebra. Boolean algebras mirror closely the structure of propositional expressions, and so arise in the study of circuit design in computer science, and more widely in many areas of mathematical logic.

Definition 5.15

A **Boolean algebra** is a set S together with operations such that the following properties hold for all a, b and c in S :

commutative laws: $a + b = b + a$, $a \cdot b = b \cdot a$
 associative laws: $(a + b) + c = a + (b + c)$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
 distributive laws: $a + (b \cdot c) = (a + b) \cdot (a + c)$, $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$
 identity laws: $a + 0 = a$, $a \cdot 1 = a$
 complement laws: $a + a' = 1$, $a \cdot a' = 0$

If B is a boolean algebra, we write $B = (S, +, \cdot, ', 0, 1)$

Note

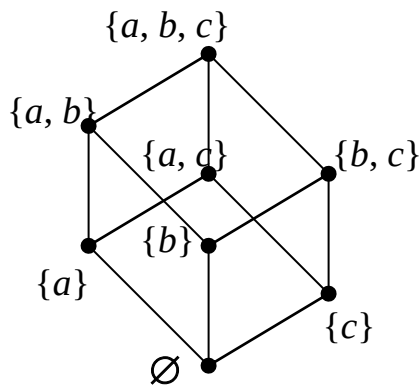
- 1) For the set of statement forms, “ $\vee$ ” and “ $\wedge$ ” play the roles of “ $+$ ” and “ $\cdot$ ”, the tautology “ t ” and contradiction “ c ” play the roles of “ 1 ” and “ 0 ”, and “ $\sim$ ” plays the roles of “ $'$ ”.
- 2) For the set of subsets of a nonempty set S , “ $\cup$ ” and “ $\cap$ ” play the role of “ $+$ ” and “ $\cdot$ ”, the set “ S ” and the empty set “ $\emptyset$ ” play the roles of “ 1 ” and “ 0 ”, and complementation “ c ” plays the roles of “ $'$ ”.

Example 5.31

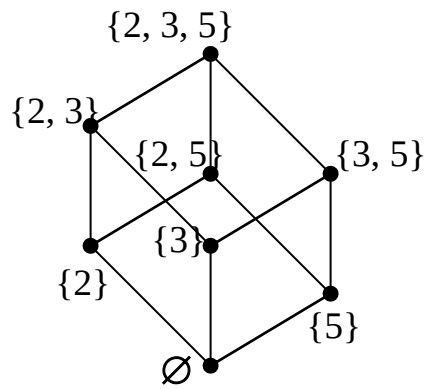
$(P(X), \cup, \cap, ^c, X, \emptyset)$ is a Boolean algebra where $P(X)$ is the power set of the non-empty set X .

Example 5.32

Let $P(S)$ be the power set of the non-empty set S . Let $S_1 = \{a, b, c\}$ and $S_2 = \{2, 3, 5\}$. The Hasse diagrams for the lattices $(P(S_1), \subseteq)$ and $(P(S_2), \subseteq)$ are shown below. Since these two posets have the same Hasse diagram, so the two lattices are isomorphic.



$(P(S_1), \subseteq)$



$(P(S_2), \subseteq)$

If $|S| = n$, then all subsets of S can be represented by sequences of 0's and 1's of length n . Suppose A is a subset of S . We assign 1 to an element that belongs to A and 0 to an element that does not belong to A .

Example 5.33

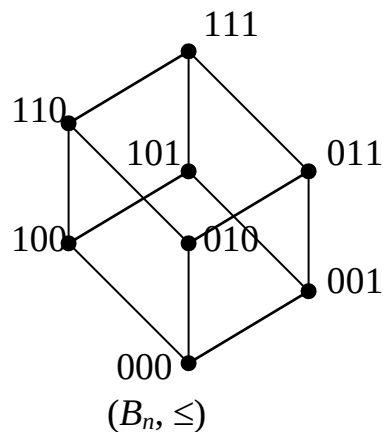
Let $S = \{1, 2, 3, 4, 5, 6\}$, $S_1 = \{1, 6\}$, $S_2 = \{1, 3, 5\}$ and $S_3 = \{2, 3, 4, 5\}$.

So, S_1 is represented by _____

S_2 is represented by _____

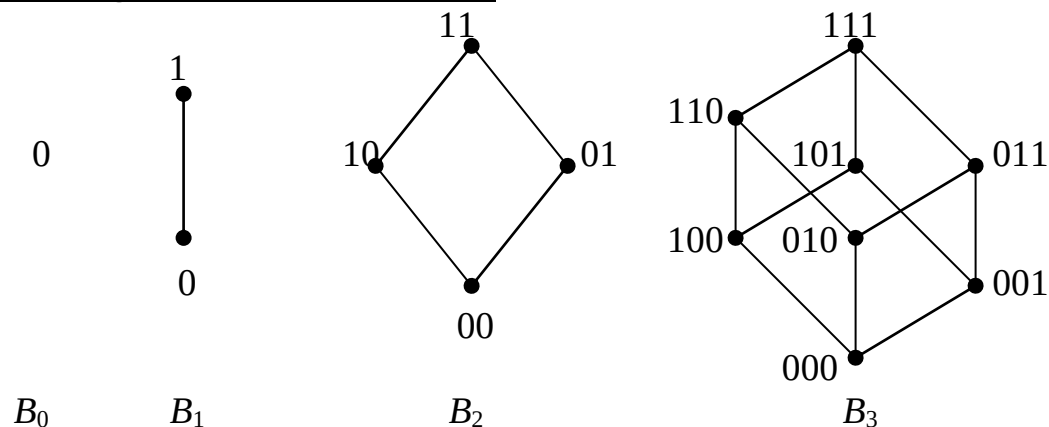
S_3 is represented by _____

Now, we can therefore label the Hasse diagram of a lattice $(P(S), \subseteq)$ by such sequences. The resulting lattice is named $\mathbf{B}_n$. In this case, we free the diagram from dependence on a particular set S and emphasize the fact that it depends only on n .



$(B_n, \leq)$

Hasse diagrams of some lattice B_n



Theorem 5.7

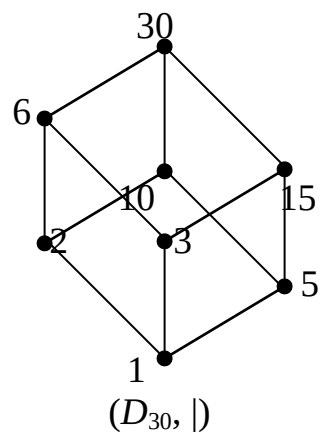
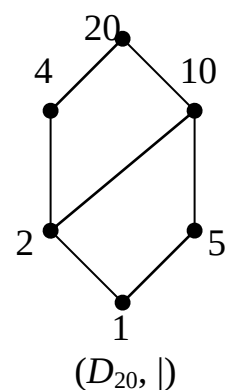
Any finite Boolean algebra is isomorphic to B_n for some nonnegative integer n .

Note

- 1) Let L be a lattice. If $|L| \neq 2^n$ for some non-negative integer n , then L is not a Boolean algebra.
- 2) Let L be a lattice. If $|L| = 2^n$ for some non-negative integer n , then L may or may not be a Boolean algebra.

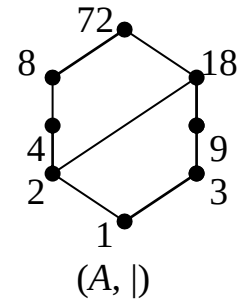
Example 5.34

Are D_{20} and D_{30} Boolean algebras?



Example 5.35

Let $A = \{1, 2, 3, 4, 8, 9, 18, 72\}$. Is $(A, |)$ a Boolean algebra?



We know that if $|L| = 2^n$ for some non-negative integer n , L may or may not be a Boolean algebra. If L is relatively small, we can compare its Hasse diagram with the Hasse diagram of B_n . However, this technique is not practical if L is large. The following theorem gives some help in solving this problem.

Theorem 5.8

Let D_n denote the set of positive integer divisors of n .

D_n is a Boolean algebra if and only if n can be written as the product of distinct prime numbers.

Note

The Unique Factorization Theorem: Any integer $n > 1$ is either prime or can be written as product of prime numbers in a way that is unique.

Example 5.36

Are D_{40} , D_{66} , D_{75} , D_{210} and D_{646} Boolean algebras?

Substitution Rule for Boolean Algebras

Theorem 5.9

Any formula involving $\cup$ or $\cap$ that holds for arbitrary subsets of a set S will continue to hold for arbitrary elements of a Boolean algebra L if $\vee$ is substituted for $\cup$ and $\cap$ for $\wedge$.

Example 5.37

If L is any Boolean algebra and x, y and z in L , then the following three properties hold.

<u>Involution Law:</u>	$(x')' = x.$	$(\overline{\overline{A}}) = A$
<u>De Morgan Law:</u>	$(x \wedge y)' = x' \vee y'.$	$(\overline{A \cap B}) = \overline{A} \cup \overline{B}$
	$(x \vee y)' = x' \wedge y'.$	$(\overline{A \cup B}) = \overline{A} \cap \overline{B}$

This is true by the Substitution rule for Boolean algebras, since we know that the corresponding formulas hold for arbitrary subsets A and B of a set S .

In a similar way, we can list other properties that must hold in any Boolean Algebra by the substitution rule.

Let $x, y, z \in L$, $A, B, C \subseteq S$ and denote the greatest and least elements of L by I and 0 , respectively.

1. $x \leq y$ if and only if $x \vee y = y$	1. $A \subseteq B$ if and only if $A \cup B = A$
2. $x \leq y$ if and only if $x \wedge y = x$	2. $A \subseteq B$ if and only if $A \cap B = A$
3. (a) $x \vee x = x$ (b) $x \wedge x = x$	3. (a) $A \cup A = A$ (b) $A \cap A = A$
4. (a) $x \vee y = y \vee x$ (b) $x \wedge y = y \wedge x$	4. (a) $A \cup B = B \cup A$ (b) $A \cap B = B \cap A$
5. (a) $x \vee (y \vee z) = (x \vee y) \vee z$ (b) $x \wedge (y \wedge z) = (x \wedge y) \wedge z$	5. (a) $A \cup (B \cup C) = (A \cup B) \cup C$ (b) $A \cap (B \cap C) = (A \cap B) \cap C$
6. (a) $x \vee (x \wedge y) = x$ (b) $x \wedge (x \vee y) = x$	6. (a) $A \cup (A \cap B) = A$ (b) $A \cap (A \cup B) = A$
7. $0 \leq x \leq I$ for all x in L .	7. $\emptyset \subseteq A \subseteq S$ for all A in $P(S)$
8. (a) $x \vee 0 = x$ (b) $x \wedge 0 = 0$	8. (a) $A \cup \emptyset = A$ (b) $A \cap \emptyset = \emptyset$
9. (a) $x \vee I = I$ (b) $x \wedge I = x$	9. (a) $A \cup S = S$ (b) $A \cap S = A$
10. (a) $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$ (b) $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$	10. (a) $A \cap (B \cup C)$ $= (A \cap B) \cup (A \cap C)$ (b) $A \cup (B \cap C)$ $= (A \cup B) \cap (A \cup C)$
11. Every element x has a unique complement x' satisfying (a) $x \vee x' = I$ (b) $x \wedge x' = 0$	11. Every element A has a unique complement $\bar{A}$ satisfying (a) $A \cup (\bar{A}) = S$ (b) $A \cap (\bar{A}) = \emptyset$
12. (a) $0' = I$ (b) $I' = 0$	12. (a) $\overline{\emptyset} = S$ (b) $\bar{S} = \emptyset$
13. $(x')' = x$	13. $\overline{(\bar{A})} = A$
14. (a) $(x \wedge y)' = x' \vee y'$ (b) $(x \vee y)' = x' \wedge y'$	14. (a) $\overline{A \cap B} = \bar{A} \cup \bar{B}$ (b) $\overline{A \cup B} = \bar{A} \cap \bar{B}$

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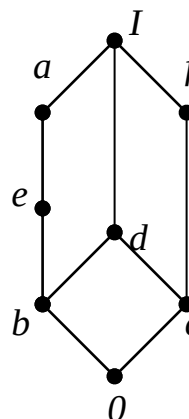
We can show that a lattice L is **not** a Boolean algebra by showing that it **does not possess** one or more of these properties.

Example 5.38

Show that the lattice whose Hasse diagram shown below is not a Boolean algebra.

Elements a and e are both complements of c ; that is, they both satisfy properties 11(a) and 11(b) with respect to the element c .

But property 11 says that such an element is unique in any Boolean algebra, thus the given lattice cannot be a Boolean algebra.



Boolean Functions as Boolean Polynomial (Simple circuit design)

Definition 5.16

Let $B = \{0, 1\}$.

A function f is called a **Boolean function of degree n** if it is a function that takes values from B_n to produce a value in B .

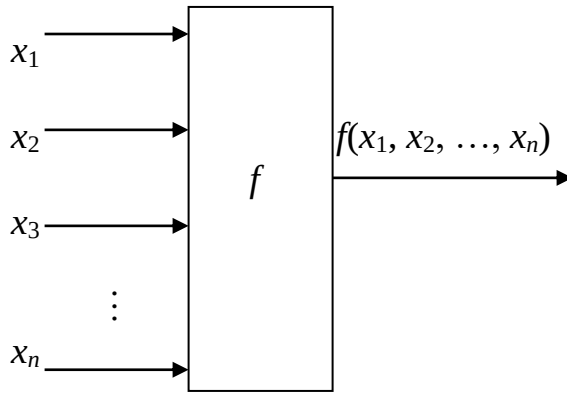
f is a function of n inputs $x_1, x_2, \dots, x_n$ of 0 or 1 and produces an output of 0 or 1.

Notation

$f: B_n \rightarrow B$.

A Boolean function may be described by a truth table showing that the values of the output for all the combinations of the values of the individual variables.

x_1	x_2	x_3	$f(x_1, x_2, \dots, x_n)$
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	



Definition 5.17

Let $x_1, x_2, \dots, x_n$ be a set of n variables.

A **Boolean polynomial** $P(x_1, x_2, \dots, x_n)$ in the variables x_k is defined recursively as follows:

- 1) The symbol 0 and 1, $x_1, x_2, \dots, x_n$ are Boolean polynomial.
- 2) If P_1 and P_2 are Boolean polynomials, then $P_1', P_1 \vee P_2$ and $P_1 \wedge P_2$ are Boolean polynomials.

Boolean polynomials are also called **Boolean expressions**.

Example 5.39

The following are examples of Boolean polynomials in variables x, y and z .

$$P_1(x, y, z) = (x \vee y) \wedge z$$

$$P_2(x, y, z) = (x \vee y') \wedge (z \wedge 1)$$

$$P_3(x, y, z) = (x \wedge (y \vee z')) \wedge ((x' \wedge 0) \vee (y' \wedge 1))$$

$$P_4(x, y, z) = (x \vee z')' \wedge (y' \vee 0)$$

$$P_5(x, y, z) = (1 \wedge (y \vee z'))$$

Example 5.40

Consider the Boolean polynomial $P(x, y, z) = (x \wedge y) \vee (x \vee (y' \wedge z))$. Construct the truth table for the Boolean function $f: B_3 \rightarrow B$ determined by this Boolean polynomial.

x	y	z	$P(x, y, z) = (x \wedge y) \vee (x \vee (y' \wedge z))$
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

How to find a Boolean expression that represents a Boolean function with given values?

If $f: B_n \rightarrow B$, we will let $S(f) = \{b \in B_n \mid f(b) = 1\}$. We then have the following result.

Theorem 5.10

Let f, f_1 and f_2 be three functions from B_n to B .

- (a) If $S(f) = S(f_1) \cup S(f_2)$, then $f(b) = f_1(b) \vee f_2(b)$ for all $b \in B$.
- (b) If $S(f) = S(f_1) \cap S(f_2)$, then $f(b) = f_1(b) \wedge f_2(b)$ for all $b \in B$.
($\vee$ and $\wedge$ are LUB and GLB, respectively, in B .)

Example 5.41

Let $f_1: B_2 \rightarrow B$ be produced by the expression $E(x, y) = x'$ and

$f_2: B_2 \rightarrow B$ be produced by the expression $E(x, y) = y'$.

The truth tables of f, f_1 and f_2 are shown as follow:

x	y	$f_1(x, y)$		x	y	$f_2(x, y)$		x	y	$f(x, y)$
0	0	1		0	0	1		0	0	1
0	1	1		0	1	0		0	1	1
1	0	0		1	0	1		1	0	1
1	1	0		1	1	0		1	1	0

Theorem 5.11

Any function for $f: B_n \rightarrow B$ for which $S(f)$ has exactly one element is produced by a Boolean expression.

The following table shows the correspondence between functions of two variables that are 1 at just one element and the Boolean expression that produces these functions.

$S(f)$	Expression producing f
$\{(0, 0)\}$	$x' \wedge y'$
$\{(0, 1)\}$	$x' \wedge y$
$\{(1, 0)\}$	$x \wedge y'$
$\{(1, 1)\}$	$x \wedge y$

Definition 5.18

If $b \in B_n$, then b is a sequence $(c_1, c_2, \dots, c_n)$ of length n , where c_k is 0 or 1.

Let E_b be the Boolean expression $\bar{x}_1 \wedge \bar{x}_2 \wedge \dots \wedge \bar{x}_n$, where

$$\begin{aligned} \bar{x}_k &= x_k \text{ when } c_k = 1 & \text{and} \\ \bar{x}_k &= x_k' \text{ when } c_k = 0. \end{aligned}$$

Such an expression is called a **minterm**.

Example 5.42

- (a) Find a minterm that equals 1 if $x = 0$ and $y = z = 1$, and equals 0 otherwise.

$$E_{(0, 1, 1)} =$$

- (b) Find a minterm that equals 1 if $x_1 = x_3 = 0$ and $x_2 = x_4 = x_5 = 1$, and equals 0 otherwise.

$$E_{(0, 1, 0, 1, 1)} =$$

By taking Boolean sum of distinct minterms we can build up a Boolean expression with a specified set of values. The sum of minterms that represents the function is called the **sum-of-products form**, or the **disjunctive normal form** of the Boolean function.

Example 5.43

Find the disjunctive normal form of the following Boolean function.

x	y	$f(x, y)$
0	0	0
0	1	1
1	0	0
1	1	1

$$E_{(x, y)} = E_{(0, 1)} \vee E_{(1, 1)} =$$

Example 5.44

Find Boolean expressions that represent the functions $f(x, y, z)$ which are given in table below.

x	y	z	$f(x, y, z)$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

The function $f: B_3 \rightarrow B$ whose truth table has $S(f) =$

So g is produced by $E(x, y, z) = (x' \wedge y \wedge z) \vee (x \wedge y \wedge z)$

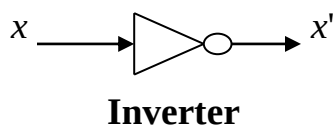
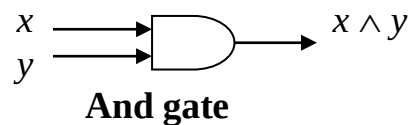
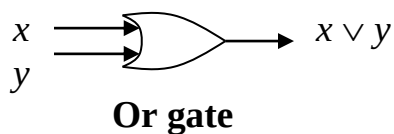
The expression $(x' \wedge y \wedge z) \vee (x \wedge y \wedge z)$ is the sum-of-products form, or the disjunctive normal form for the function. However, is not the simplest Boolean expression that produces f .

Using properties of Boolean algebras, we have

Thus, f is also produced by the simple expression $y \wedge z$.

Logic Diagram

Boolean polynomials can also be written in a graphical or schematic way as follows:



By repeated use of $\wedge$, $\vee$, $'$, we can make a schematic form to represent any Boolean polynomial. Such diagram is called **logic diagram** for the polynomial.

Example 5.45

Let $p(x, y, z) = (x \wedge y) \vee (y \wedge z')$. Construct the truth table for the corresponding function $f: B_3 \rightarrow B$. Draw the logic diagram for p .

x	y	z	z'	$x \wedge y$	$y \wedge z'$	$f(x, y, z)$
1	1	1				
1	1	0				
1	0	1				
1	0	0				
0	1	1				
0	1	0				
0	0	1				
0	0	0				